Ethereum Rollup Settlement Take Rate After Dencun

Validated working paper source for the L2-to-L1 rent analysis release candidate

autonomous-agentic-research-swarm

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1 Abstract

This paper documents the current release-candidate estimate of Ethereum rollup Settlement Take Rate (STR), the share of rollup fee revenue paid back to Ethereum L1 as settlement rent. The evidence bundle uses validated local artifacts as of `2026-04-09` and spans `14` rollups over `1,559` daily observations from `2022-01-01` through `2026-04-08` (autonomous-agentic-research-swarm repository 2026e, 2026a). On the locked regime table, mean ecosystem STR is `69.14%` before Dencun and `11.68%` after Dencun, with a lower `8.98%` mean during post-Dencun blob-fee-floor runs (autonomous-agentic-research-swarm repository 2026f, 2026i). The manuscript is intentionally downstream of repo contracts and validation outputs: it reports the current validated surface, documents provenance, and records remaining release caveats instead of redefining metrics in prose.

2 Research Question

The repository's locked project question is whether Ethereum L1's settlement take rate on rollup economics is rising, stable, or decaying as L2 usage scales (autonomous-agentic-research-swarm repository 2026e). The primary metric is Settlement Take Rate (STR), defined in the protocol as

```
STR_t = (sum_i RentPaid_{i,t}) / (sum_i L2Fees_{i,t})
```

where `RentPaid` is the authoritative on-chain estimate of rollup payments to Ethereum L1 and `L2Fees` comes from the locked vendor denominator surface (autonomous-agentic-research-swarm repository 2026f). The current validated evidence supports a specific descriptive answer: ecosystem STR decays sharply after the Dencun boundary on `2024-03-13`, even though rollup fee activity remains substantial (autonomous-agentic-research-swarm repository 2026f, 2026i).

3 Data And Protocol

This draft uses only durable local artifacts already produced upstream of the paper layer. The core empirical inputs are the canonical rollup panel, the daily L1 rent decomposition, the validation bundle, and the W6 release figures/table (autonomous-agentic-research-swarm repository 2026a, 2026c, 2026b, 2026h, 2026d, 2026i).

Four protocol rules matter most for interpreting the paper (autonomous-agentic-research-swarm repository 2026f):

- STR is computed daily in ETH-native units.
- A panel row exists only when both `l2_fees_eth` and `rent_paid_eth` are present.
- Dates on or after `2024-03-13` are post-Dencun.
- Blob-fee-floor runs are post-Dencun stretches of at least `7` consecutive days where blob base fee stays within `1.05 x` the post-Dencun minimum.

On the validated `2026-04-09` surface, the authoritative panel contains `12,434` rollup-day rows across `14` rollups and `1,559` dates, while the decomposition covers the full `2022-01-01` through `2026-04-08` daily window (autonomous-agentic-research-swarm repository 2026a, 2026c). The manuscript therefore treats the release figures and regime table as direct downstream artifacts of the locked analysis path rather than restating results from untracked calculations.

4 Validation

All three upstream validation reports currently pass on the same `2026-04-09` as-of surface (autonomous-agentic-research-swarm repository 2026a, 2026c, 2026b). The locked checks establish three distinct claims.

First, the authoritative panel and the rent-component surface satisfy schema, primary-key, and non-null requirements, and the component identities reconcile exactly to canonical `rent_paid_eth` under both tx-family and fee-class decompositions (autonomous-agentic-research-swarm repository 2026a). Second, daily summed components match `daily_l1_rent_decomposition.l1_total_rent_eth` within the locked tolerance over all `1,559` dates (autonomous-agentic-research-swarm repository 2026c). Third, canonical and vendor panels share full matched-key coverage, show no material unexplained rollup divergences, and differ by only about `0.02%` in aggregate rent over the full matched slice (autonomous-agentic-research-swarm repository 2026b).

The validation bundle still carries one documented non-gating residual worth keeping visible in release prose: in `2025-05`, authoritative monthly rent is `148.42 ETH` versus `166.78 ETH` in the vendor benchmark, a `12.37%` gap on a low-rent month (autonomous-agentic-research-swarm repository 2026b, 2026g). Under the locked benchmark policy, that remains diagnostic context rather than grounds to overwrite canonical rent or block the paper source.

5 Results

Figure Figure 1 shows the ecosystem time series built from the validated panel surface. The series makes the main qualitative result easy to see: pre-Dencun STR is frequently elevated, while post-Dencun STR is structurally lower even when L2 fees remain meaningful (autonomous-agentic-research-swarm repository 2026h). The accompanying regime table quantifies that break. Mean ecosystem STR falls from `69.14%` before Dencun to `11.68%` after Dencun, while mean daily rent drops from `147.014 ETH` to `19.064 ETH` (autonomous-agentic-research-swarm repository 2026i).

Figure Figure 2 focuses on the post-Dencun sample and shades the protocol-defined blob-fee-floor runs (autonomous-agentic-research-swarm repository 2026f, 2026d). The table shows that mean STR is `8.98%` during blob-fee-floor periods versus `12.77%` in the rest of the post-Dencun sample (autonomous-agentic-research-swarm repository 2026i). On the current validated surface, the simplest answer to the project's primary question is therefore not "rising" or "stable" but "decaying after the Dencun regime change," with the cheapest blob-fee conditions coinciding with the lowest observed ecosystem rent take rates.

5.1 Regime Summary Table

The locked W6 regime table is included directly from the durable table surface so the manuscript and release bundle reference the same artifact.

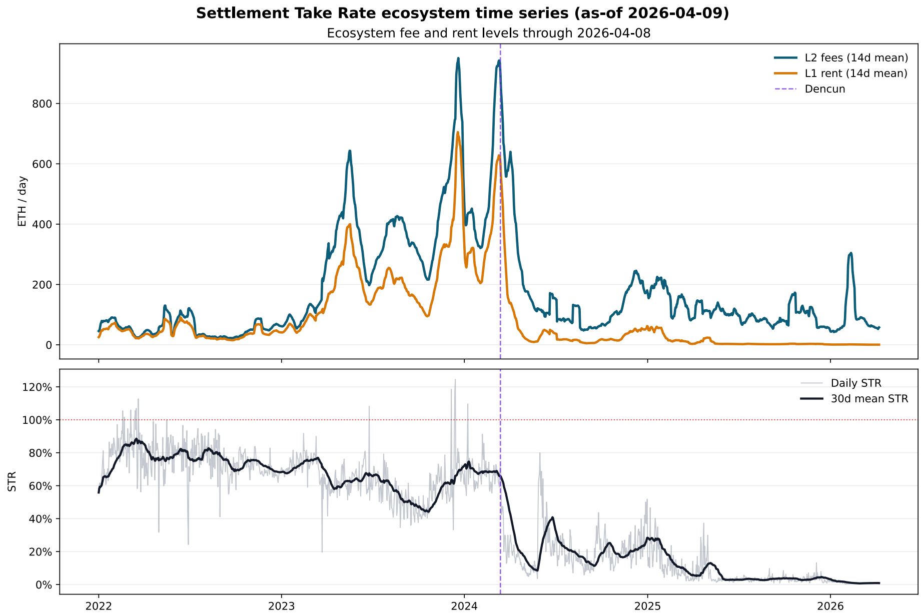

Figure 1: Settlement Take Rate ecosystem time series through 2026-04-08. The released figure overlays daily STR with smoothed L2 fee and L1 rent levels, and marks the Dencun boundary at 2024-03-13.

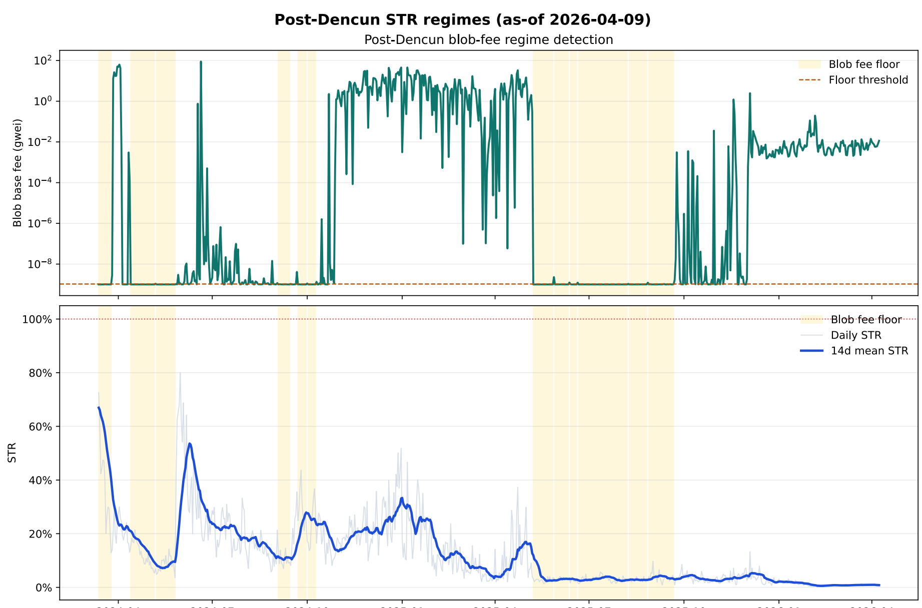

Figure 2: Post-Dencun STR regime figure with blob-fee-floor runs shaded according to the locked protocol rule.

regime_id	regime	start_date_utc	end_date_utc	days	mean_l2_fees	mean_rent_paid	mean_str_pct	median_str_pct	p90_str_pct	mean_blob_base_fee_gwei
full_sample	Full sample	2022-01-01	2026-04-08	1559	181.399	84.886	41.24%	44.59%	79.85%	2.250096
pre_dencun	Pre-Dencun	2022-01-01	2024-03-12	802	229.556	147.014	69.14%	69.36%	84.97%	
post_dencun	Post-Dencun	2024-03-13	2026-04-08	757	130.379	19.064	11.68%	6.55%	27.64%	2.250096
post_dencun_blob_floor	Blob fee floor	2024-03-13	2025-09-21	217	128.737	18.740	8.98%	3.93%	24.06%	0.000000
post_dencun_non_floor	Post-Dencun ex floor	2024-03-26	2026-04-08	540	131.038	19.194	12.77%	9.09%	28.61%	3.154302

6 Provenance And Limitations

This manuscript is downstream of the locked repository control plane, not a parallel analysis channel. Protocol definitions come from `docs/protocol.md`, project-level release requirements come from `contracts/project.yaml`, and the empirical claims quoted here are limited to validated report artifacts and their provenance manifests (autonomous-agentic-research-swarm repository 2026f, 2026e, 2026a, 2026c).

Two limitations matter for interpretation. First, the authoritative panel is intentionally denominator-keyed: rollup-days missing vendor fee denominators are omitted from STR aggregation even when rent-component data exists. The current validated surface therefore has `12,434` panel rows but `12,563` rent-component rows, and the component-to-decomposition identity is the correct full-rent coherence check (autonomous-agentic-research-swarm repository 2026a, 2026g). Second, this task owns manuscript source only. Final rendered artifacts, release manifests, and catalog updates remain Operator-owned T080 surfaces.

Within those limits, the paper source is release-aligned: it uses the locked filenames, cites only durable local artifacts, and can be rendered by Quarto without inventing new evidence or changing scientific definitions.

autonomous-agentic-research-swarm repository. 2026a. *Canonical Rollup Panel Validation.*

autonomous-agentic-research-swarm repository. 2026b. *Cross-Source Reconciliation.*

autonomous-agentic-research-swarm repository. 2026c. *L1 Rent Decomposition Validation.*

autonomous-agentic-research-swarm repository. 2026d. *Post-Dencun STR Regimes.*

autonomous-agentic-research-swarm repository. 2026e. *Project Contract: L2-to-L1 Rent Analysis.*

autonomous-agentic-research-swarm repository. 2026f. *Protocol Lock (Phase 0).*

autonomous-agentic-research-swarm repository. 2026g. *Release Output Caveats for T070.*

autonomous-agentic-research-swarm repository. 2026h. *Settlement Take Rate Ecosystem Time Series.*

autonomous-agentic-research-swarm repository. 2026i. *STR Regime Summary Table.*